ADR-0003: Simulated contestant behavior and the tiebreaker rule

Status: accepted. Supersedes ADR-0002. Its accuracy, correlation, leader-cover and Daily Double values are superseded by ADR-0004 (modern-era calibration). Date: 2026-10-13

Context

Two defects in the Tier 2 and Daily Double engines surfaced on 2026-10-13.

1. **Wrong tie rule.** The engines counted a tie for first as a full win, on the grounds that co-champions both return. That rule ended with the 24 November 2014 episode. Since then, a tie for first after Final Jeopardy is settled by a tiebreaker clue, and only one player returns (Wikipedia, "Jeopardy!", rules section; date from the Jeopardy! History Wiki, Season 31, show #6946). The change followed the October 2014 co-champion ties; it was formally announced in 2016, and the first regular-play tiebreaker came in 2018 (McNear, The Ringer, 2021). The visible symptom: for a leader whose cover bet was $6,001, Smart Bet recommended $6,000 — a bet that only ties second place if they double up. Under the real rule that tie is worth about half a win.
2. **Placeholder opponents.** ADR-0002 shipped a uniform split over each opponent's own Tier 1 breakpoints, at 50% accuracy, independent across players. It was labeled as a placeholder. Tesauro et al. (2013, JAIR 47:205-251) publish enough about real contestant behavior to do better.

Decision

All values live in one configuration block in `engine/behavior.py`, mirrored in the `BEHAVIOR` object in `app/index.html`.

Scoring. A player's result is a *win share*: 1 for an outright win, 1/k for a k-way tie for first (equal tiebreaker odds, assumed), 0 otherwise. Nobody wins if the top score is $0 or less.

Opponent wagers. Each opponent is assigned a role by score: leader (A), second (B), third (C).

Role	Bet types and weights	Source
B	all-in 26%, keep C out (bet < B-2C, or cover 2C-B+1) 27%, overtake A (A-B+1) 15%, two-thirds (3B-2A-1) 8%, random 24%	PUBLISHED for the regime B ≥ ¾A and B ≥ 2C (T13 §2.3). The "cover 2C-B+1" branch is an ASSUMED extension (it applies only when B < 2C, outside the published regime). ASSUMED for the other five regimes.
A, not locked	cover 2B-A+1 75%, tie-cover 2B-A 10%, all-in 5%, random 10%	ASSUMED. T13 §1.1 says A "usually" covers, "frequently" 2B-A+1, with no split.

Role	Bet types and weights	Source
A, lockout (A > 2B)	$0 50%, largest safe bet A-2B-1 50%	ASSUMED.
C	even spread over the random grid	PUBLISHED that T13 found no evidence that C's bets track the game state (T13 §2.3 note 5). The even spread is ASSUMED.

"Random" is an equal spread over 0, ¼, ½, ¾ and all of the score. When a bet type does not exist in a state (for example, no third player to keep out), its weight is shared out in proportion among the others.

Accuracy. Opponents answer Final Jeopardy right 50% of the time (PUBLISHED, average contestants, T13 §2.3; champions 60%). Right/wrong outcomes are correlated across all players at 0.3 (PUBLISHED, T13 §2.3), modeled as a one-factor Gaussian copula. We read 0.3 as the binary correlation at 50% accuracy and convert it with Sheppard's formula to a latent correlation $r = \sin(0.3\pi/2) \approx 0.454$. The user is inside the same copula: a hard clue is hard for everyone.

Daily Double. Future Daily Doubles are answered right 64% of the time (PUBLISHED, average contestants, T13 §2.2). Inside a rollout, opponents' Final Jeopardy wagers are drawn from the model above.

Computation. Tier 2 is exact: it integrates over the shared factor (trapezoid rule, step 0.25 on [-6, 6]) and enumerates every wager and right/wrong branch. The DD engine samples the same model.

Consequences

Positive:

- Smart Bet no longer recommends a tie-only bet when one more dollar wins.
- Every opponent number is either cited or labeled as an assumption, in one place.
- Correlated outcomes capture a real effect: when you miss, a rival is more likely to have missed too.

Negative:

- Five of the six B regimes, all of A's split and C's shape are our assumptions, not fitted. The published B weights came from groups of "only a few hundred samples" each (T13 §2.3), so even those carry wide error bars.
- The data are from the mid-1990s to 2012 at the latest. Contestant behavior has likely changed since; non-stationarity is not modeled.
- Equal tiebreaker odds is an assumption; buzzer skill decides the clue.
- The paper defines rho on the binary outcomes (T13 §1.1 Glossary: rho = P(RR) + P(WW) - P(RW) - P(WR)) and generates it by thresholding correlated normals (§2.3). This equals the phi correlation at 50% accuracy; at other accuracies the Sheppard conversion is approximate.
- Win percentages from any earlier version are not comparable: they scored ties as wins against a different opponent model.
- The app's Daily Double calculation takes about twice as long, because each rollout now simulates opponent wagers.